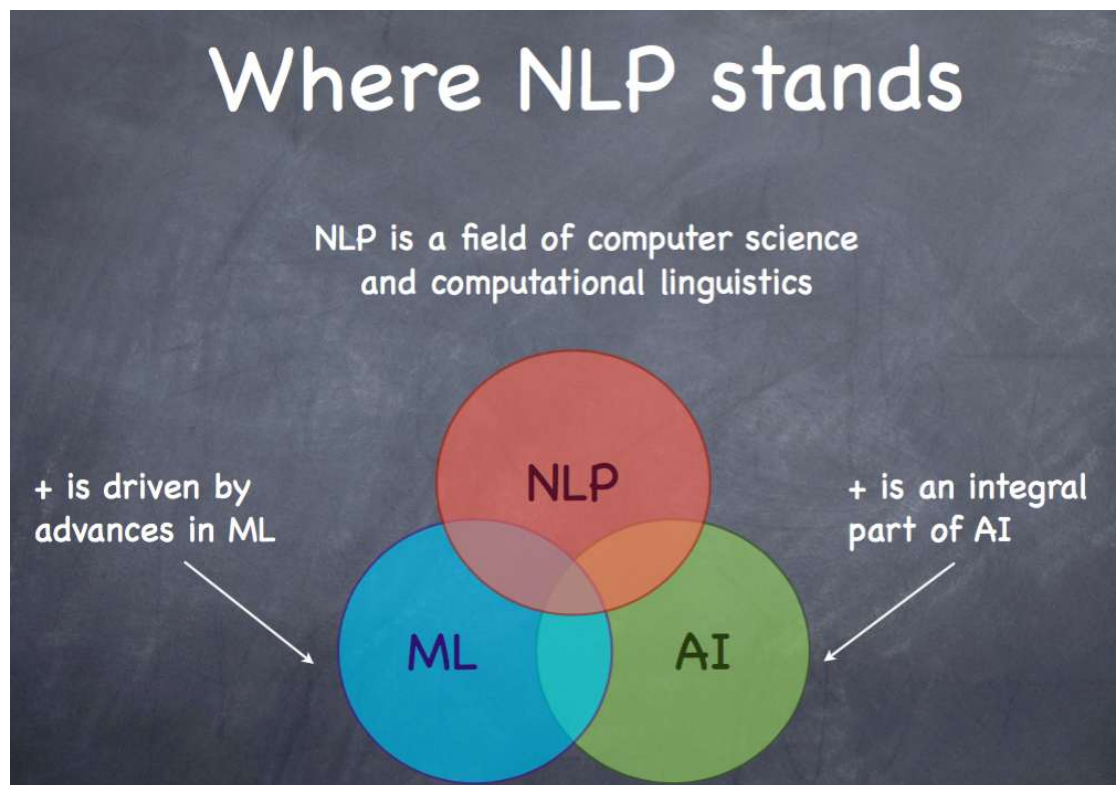


Introduction to Natural Language processing

Natural language processing (NLP): Subfield of linguistics, computer science, and information engineering concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyze large amounts of natural language data.



What is it all about?

What's that all about

Let's start with the basics:



we speak



we read



we write

all that using language

But that's not all...



- we think of the world around us
- we dream
- we make decisions and plans
- all in natural language, i.e. in words

What are the raw materials for NLP revolution?

1. Data: There is a huge amount of human knowledge that has been digitized.

2. Compute: Our personal computers are very powerful, and it has never been easier to rent more compute power.

3. Algorithms: We use a combination of Computer Science, Machine Learning, Deep Learning, and linguist techniques.

Why NLP is hard?

It's the nature of the human language that makes NLP difficult.

Some examples:

1. Ambiguity: One word/sentence can have different meanings.



2. Spelling Errors: There are some words that are usually misspelled by people.



Top 10 Misspelled Words

separate



seperate



calendar



calender



unnecessary



unecessary



questionnaire



questionaire



accidentally



accidentaly



necessary



neccessary



surprise



suprise



embarrass



embarass



accommodation



acomodation



receive



recieve



3. Phonetics and Phonology: Different words that have the same pronunciation.

- there / their
- plane / plain
- ice cream / I scream

4. Order of Context: "Dog bites man" and "Man bites dog" have vastly different meanings

In ML terms, why is NLP hard?

1. Input: Large, discrete, open-ended input state spaces

2. Output: Variable length output spaces

3. Algorithms: Recycled from other fields

Top Used Text File Types

- CSV
- Text Files
- JSON
- SQL
- HTML
- PDF
- DOCX

NLP Libraries

There are many popular NLP libraries that we will cover in the next labs. In this lab, we will get introduced to NLTK.

NLTK

Natural Language Toolkit (nltk) is a Python package for NLP <https://www.nltk.org/>

What are Corpus / Corpora?

a collection of written texts, especially the entire works of a particular author or a body of writing on a particular subject.

```
In [1]: !pip install nltk
```

```

Collecting nltk
  Using cached nltk-3.10.3-py3-none-any.whl.metadata (3.2 kB)
Collecting defusedxml (from nltk)
  Using cached defusedxml-0.7.1-py2.py3-none-any.whl.metadata (32 kB)
Collecting click (from nltk)
  Using cached click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting joblib (from nltk)
  Using cached joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting regex<=2021.8.3 (from nltk)
  Downloading regex-2026.9.29-cp313-cp313-win_amd64.whl.metadata (41 kB)
Collecting tqdm (from nltk)
  Using cached tqdm-4.70.1-py3-none-any.whl.metadata (57 kB)
Collecting cloudpickle>=3.0 (from joblib->nltk)
  Using cached cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Collecting colorama (from tqdm->nltk)
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Using cached nltk-3.10.3-py3-none-any.whl (1.8 MB)
Downloading regex-2026.9.29-cp313-cp313-win_amd64.whl (280 kB)
Using cached click-8.5.0-py3-none-any.whl (125 kB)
Using cached defusedxml-0.7.1-py2.py3-none-any.whl (25 kB)
Using cached joblib-1.6.0-py3-none-any.whl (306 kB)
Using cached cloudpickle-3.1.2-py3-none-any.whl (22 kB)
Using cached tqdm-4.70.1-py3-none-any.whl (80 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: regex, defusedxml, colorama, cloudpickle, click, t
qdm, joblib, nltk

```

```

----- 4/8 [click]
----- 6/8 [joblib]
----- 6/8 [joblib]
----- 7/8 [nltk]
----- 7/8 [nltk]
----- 7/8 [nltk]
----- 7/8 [nltk]
----- 7/8 [nltk]
----- 7/8 [nltk]
----- 7/8 [nltk]
----- 8/8 [nltk]

```

Successfully installed click-8.5.0 cloudpickle-3.1.2 colorama-0.4.6 defusedxml-0.7.1 joblib-1.6.0 nltk-3.10.3 regex-2026.9.29 tqdm-4.70.1

```
In [1]: import nltk
print(nltk.__version__)
```

3.10.3

Arabic Libraries (Extra)

Camel-Tools: <https://camel-tools.readthedocs.io/en/latest/>

PyArabic: <https://pyarabic.readthedocs.io/ar/latest/>

Farasa: <https://farasa.qcri.org/>

NLP Applications

1. Information Extraction

The automated retrieval of specific information related to a selected topic from a body or bodies of text.

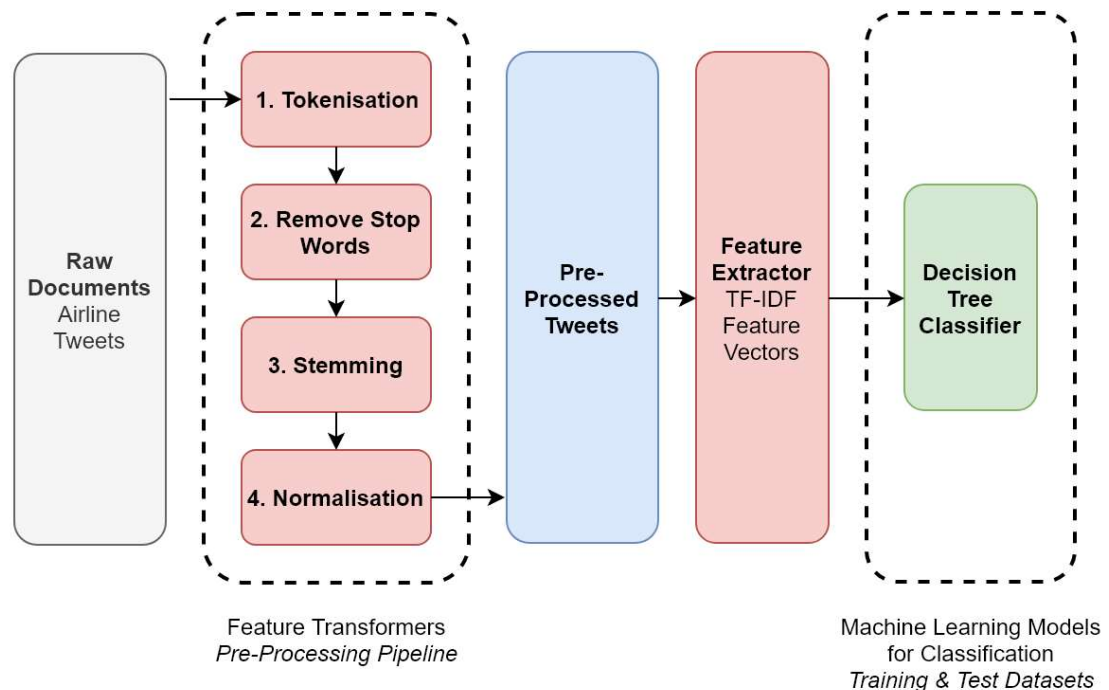
2. Machine Translation

The process of using artificial intelligence to automatically translate text from one language to another without human involvement.

3. Sentiment Analysis

Identifying sentiments and opinions stated in a text.

Example of an NLP Pipeline



1. Text Acquisition:

Obtain the text data from a source, such as a file, a web page, or a database.

2. Text Preprocessing:

- **Tokenization:** Split the text into individual words or tokens.
- **Lowercasing:** Convert all tokens to lowercase to ensure uniformity.
- **Removing Punctuation:** Strip away punctuation marks from tokens.
- **Stopword Removal:** Eliminate common words (e.g., "and," "the," "is") that may not carry significant meaning.
- **Stemming:** Removes prefixes/suffixes to get a root form. (Running -> run)
- **Lemmatization:** Converts word to base/dictionary form (lemma). (better -> good).

3. Feature Extraction:

- **Bag-of-Words (BoW):** Create a vector representation of the text using the frequency of each token in the document.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Assign weights to tokens based on their importance in the document relative to the entire corpus.

4. Model Building:

Choose a machine learning model for sentiment analysis, such as a Naive Bayes classifier, Support Vector Machine (SVM), or a neural network.

Task#1:

Students should work in groups and search online for one of the different types of datasets. • CSV • Text Files • JSON • SQL • HTML • PDF • DOCX

Task#2:

Download and open the dataset.

```
In [4]: %pip install kagglehub
```

Collecting kagglehub

Downloading kagglehub-1.0.2-py3-none-any.whl.metadata (40 kB)

Collecting kagglesdk<1.0,>=0.1.22 (from kagglehub)

Downloading kagglesdk-0.1.37-py3-none-any.whl.metadata (13 kB)

Requirement already satisfied: packaging in C:\Users\fjahl\AppData\Roaming\Python\Python314\site-packages (from kagglehub) (26.3)

Collecting pyyaml (from kagglehub)

Using cached pyyaml-6.0.3-cp314-cp314-win_amd64.whl.metadata (2.4 kB)

Collecting requests (from kagglehub)

Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)

Requirement already satisfied: tqdm in c:\Users\fjahl\AppData\Local\Python\python-core-3.14-64\Lib\site-packages (from kagglehub) (4.70.1)

Collecting protobuf (from kagglesdk<1.0,>=0.1.22->kagglehub)

Downloading protobuf-7.36.2-cp310-abi3-win_amd64.whl.metadata (595 bytes)

Collecting charset_normalizer<4,>=2 (from requests->kagglehub)

Downloading charset_normalizer-3.5.2-cp314-cp314-win_amd64.whl.metadata (47 kB)

Collecting idna<4,>=2.5 (from requests->kagglehub)

Using cached idna-3.20-py3-none-any.whl.metadata (7.2 kB)

Collecting urllib3<3,>=1.26 (from requests->kagglehub)

Using cached urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)

Collecting certifi>=2023.5.7 (from requests->kagglehub)

Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)

Requirement already satisfied: colorama in C:\Users\fjahl\AppData\Roaming\Python\Python314\site-packages (from tqdm->kagglehub) (0.4.6)

Downloading kagglehub-1.0.2-py3-none-any.whl (70 kB)

Downloading kagglesdk-0.1.37-py3-none-any.whl (262 kB)

Downloading protobuf-7.36.2-cp310-abi3-win_amd64.whl (456 kB)

Using cached pyyaml-6.0.3-cp314-cp314-win_amd64.whl (156 kB)

Using cached requests-2.34.2-py3-none-any.whl (73 kB)

Downloading charset_normalizer-3.5.2-cp314-cp314-win_amd64.whl (210 kB)

Using cached idna-3.20-py3-none-any.whl (69 kB)

Using cached urllib3-2.8.0-py3-none-any.whl (135 kB)

Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)

Installing collected packages: urllib3, pyyaml, protobuf, idna, charset_normalizer, certifi, requests, kagglesdk, kagglehub

```

----- 2/9 [protobuf]
----- 2/9 [protobuf]
----- 6/9 [requests]
----- 7/9 [kagglesdk]
----- 7/9 [kagglesdk]
----- 7/9 [kagglesdk]
----- 9/9 [kagglehub]

```

Successfully installed certifi-2026.7.22 charset_normalizer-3.5.2 idna-3.20 kagglehub-1.0.2 kagglesdk-0.1.37 protobuf-7.36.2 pyyaml-6.0.3 requests-2.34.2 urllib3-2.8.0

Note: you may need to restart the kernel to use updated packages.

WARNING: The script idna.exe is installed in 'c:\Users\fjahl\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script normalizer.exe is installed in 'c:\Users\fjahl\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

[notice] A new release of pip is available: 26.1.2 -> 26.2.1

[notice] To update, run: python.exe -m pip install --upgrade pip

```
In [4]: import kagglehub

# IMDB Dataset of 50K Movie Reviews
path = kagglehub.dataset_download("lakshmi25npathi/imdb-dataset-of-50k-movie-rev

print("Path to dataset files:", path)
```

c:\Users\fjah1\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html

from .autonotebook import tqdm as notebook_tqdm
Downloading to C:\Users\fjah1\.cache\kagglehub\datasets\lakshmi25npathi\imdb-dataset-of-50k-movie-reviews\1.archive...

100%|██████████| 25.7M/25.7M [00:40<00:00, 668kB/s]

Extracting files...

Path to dataset files: C:\Users\fjah1\.cache\kagglehub\datasets\lakshmi25npathi\imdb-dataset-of-50k-movie-reviews\versions\1

```
In [23]: import pandas as pd

# Load the CSV file in a dataframe
df = pd.read_csv("IMDB Dataset.csv")

# Show the first few rows
df.head()
```

```
Out[23]:
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

```
In [24]: df.describe()
```

```
Out[24]:
```

	review	sentiment
count	50000	50000
unique	49582	2
top	Loved today's show!!! It was a variety and not...	positive
freq	5	25000

```
In [25]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   review      50000 non-null   str
1   sentiment   50000 non-null   str
dtypes: str(2)
memory usage: 781.4 KB
```

```
In [ ]: # Check class distribution
        print(df['sentiment'].value_counts())
```

```
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```